Introduction

The purpose of this document is to evaluate potential security weaknesses and implement security controls within the mobile app's infrastructure.

Scope of Work

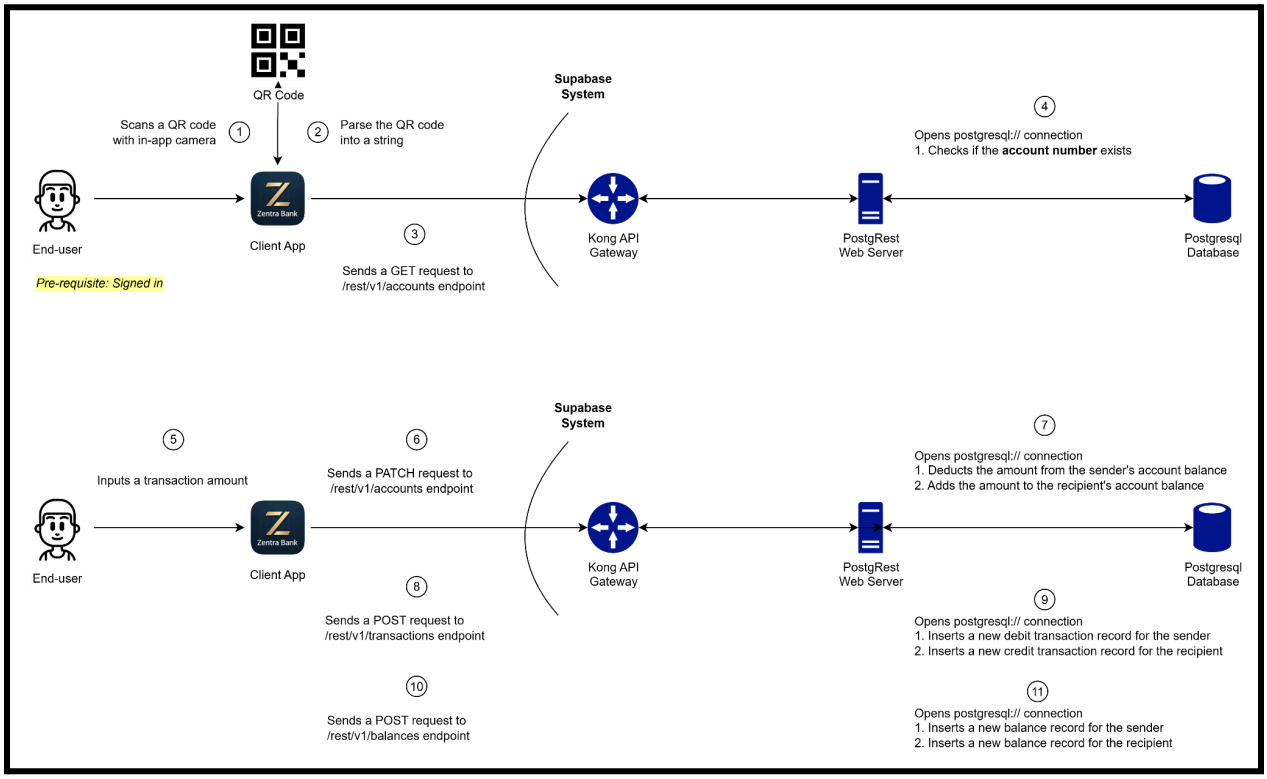
This exercise focuses on 1 user story within the Zentra Bank mobile app:

1. As a user, I want to **transfer funds using "PayNow" thru QR code** so that I can have another alternative to send money to friends, family, or service providers

Out of Scope

This exercise strictly focuses on the mobile application and its functionalities. Any components outside the mobile app, which includes but not limited to REST APIs, server infrastructure, remote database, and external services, are not within the scope.

User story 5



The data flow for this user story is as follows:

"As a user, I want to transfer funds using "PayNow" thru QR code so that I can have another alternative to send money to friends, family, or service providers."

1. The End-user scans a QR code with the in-app camera
2. The Client App (CA) parses the QR code into an account number string and sends a check request via REST API to the Web Server (WS) to verify whether it exists
3. The Web Server (WS) checks 1) if the account number exists against the `accounts` table in the Database (DB) via Postgres connection

```
GET /rest/v1/accounts?select=*&account_number=eq.00177714179 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
accept: application/json
accept-profile: public
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9
2MjQsImV4cCI6MjA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc
2F1dGgvdjEiLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhd
TY4OTIyMzIsImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZ
l19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92Z
mMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0Z
nRpbWVzdGFtcCI6MTc1Njg5MjIzMn1dLCJzZXNzaW9uX2lkIjoiNGYzYmJiOWMtNzE0OC00MGEyL
4_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
x-client-info: supabase-js-react-native/2.49.5-next.1
```

4. The End-user inputs a transaction amount

5. To complete the transfer, the Client App (CA) first sends a request to update accounts' balances via REST API to the Web Server (WS) to 1) deduct the amount from the sender's account balance and 2) add the amount to the recipient's account balance

6. The Web Server (WS) makes an update query to the **`accounts`** table in the Database (DB) via Postgres connection. The Database (DB) then 1) deducts the amount from the sender's account balance and 2) adds the amount to the recipient's account balance

```
PATCH /rest/v1/accounts?account_id=eq.4fc4ca5e-09d2-433c-9321-add4c91aa6a2 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cH
2MjQsImV4cCI6MjA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRw
2F1dGgvdjEiLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhdWQiOiJhdXRo
TY4OTIyMzIsImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZGF0YSI6eyJw
l19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92ZXJpZmllZCI6
mMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0ZWQiLCJhYWwi
nRpbWVzdGFtcCI6MTc1Njg5MjIzMn1dLCJzZXNzaW9uX2lkIjoiNGYzYmJiOWMtNzE0OC00MGEyLTljNTctMzc1
4_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 21

{"current_balance":2}
```

```
PATCH /rest/v1/accounts?account_id=eq.15872f95-e28a-4dc6-9bdc-6ffc22e6e0b2 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cHN
2MjQsImV4cCI6MjA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRw
2FldGgvdjEiLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhdWQiOiJhdXRo
TY4OTIyMzIsImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZGF0YSI6eyJw
l19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGVzdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92ZXJpZmllZCI6
mMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0ZWQiLCJhYWwi
nRpbWVzdGFtcCI6MTc1Njg5MjIzMn1dLCJzZXNzaW9uX2lkIjoiNGYzYmJiOWMtNzE0OC00MGEyLTljNTctMzc1
4_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 23

{"current_balance":18}
```


7. The Client App (CA) will also send a request to insert new transaction records via REST API to the Web Server (WS) for audit trail

8. The Web Server (WS) makes an insert query to the **`transactions`** table in the Database (DB) via Postgres connection. The Database (DB) then inserts 1) a debit transaction record for the sender and 2) a credit transaction record for the recipient


```
POST /rest/v1/transactions?columns="account_id","booking_datetime","amount","currency","credit_debit","description","transaction_type","status" HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cHNwYXVwYmFoIiwicm9sZSI6ImFub24iLCJpYXQiOjE3NDg0ODA2MjQsImV4cCI6MjA2NDA1N
BmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRwczovL3ZvbW9oY2NtYXB2cHNwYXVwYmFoLnN1cGFiYXNlLmNvL2F1dGgvdjEiLCJzdWIiOiI
C00MjRlYWYxNjExZDcifSwicm9sZSI6ImF1dGhlbnRpY2F0ZWQiLCJleHAiOjE3NTY4OTU4MzIsImlhdCI6MTc1Njg5MjIzMiwidXNlcl9tZXRhZGF0YSI6eyJlbWFpbCI6InRlc3RuaWNAbG1haWwuY29tIn1
tcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 430

[{
  "account_id": "4fc4ca5e-09d2-433c-9321-add4c91aa6a2",
  "booking_datetime": "2025-09-03T09:49:48.381Z",
  "amount": 1,
  "currency": "SGD",
  "credit_debit": "Debit",
  "description": "Transfer to 00177714179",
  "transaction_type": "Online Transfer",
  "status": "Booked"
},{
  "account_id": "15872f95-e28a-4dc6-9bdc-6ffc22e6e0b2",
  "booking_datetime": "2025-09-03T09:49:48.381Z",
  "amount": 1,
  "currency": "SGD",
  "credit_debit": "Credit",
  "description": "Transfer from 00139540222",
  "transaction_type": "Online Transfer",
  "status": "Booked"
}]
```


9. The Client App (CA) will also send a request to insert new balance records via REST API to the Web Server (WS) for audit trail

10. The Web Server (WS) makes an insert query to the **`balances`** table in the Database (DB) via Postgres connection. The Database (DB) then inserts a new balance record for 1) the sender and 2) the recipient

```
POST /rest/v1/balances?columns="account_id","balance_type","amount","currency","credit_debit","reference_date" HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW9oY2NtYXB2cHNwYXVwYmFoIiwicm9sZSI6ImFub24iLCJpY
jA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJpc3MiOiJodHRwczovL3ZvbW9oY2NtYXB2cHNwYXVwYmFoLnN
zdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZSI6ImFldGhlbnRpY2F0ZWQiLCJleHAiOjE3NTY4OTU4MzIsImlhdCI6M
GF0YSI6eyJlbWFpbCI6InRlc3RuaWNAbGlhaWwuY29tIn19.XmKw04_LDBVWcrYCogGWf4BC7gjBtcqyMWL3VIo8BR4
content-profile: public
content-type: application/json
prefer:
x-client-info: supabase-js-react-native/2.49.5-next.1
Content-Length: 373

[{
  "account_id": "4fc4ca5e-09d2-433c-9321-add4c91aa6a2",
  "balance_type": "InterimAvailable",
  "amount": 2,
  "currency": "SGD",
  "credit_debit": "Debit",
  "reference_date": "2025-09-03T09:49:48.514Z"
},{
  "account_id": "15872f95-e28a-4dc6-9bdc-6ffc22e6e0b2",
  "balance_type": "InterimAvailable",
  "amount": 18,
  "currency": "SGD",
  "credit_debit": "Credit",
  "reference_date": "2025-09-03T09:49:48.514Z"
}]
```

Bonus user story

Insert diagram here

The data flow for this user story is as follows:

“As a user, I want to generate my own QR code so others can have an alternative means to send me money easily.”

1. The End-user generates a QR code by clicking a button
2. The Client App (CA) sends a request via REST API to the Web Server (WS) to retrieve the user’s account number
3. The Web Server (WS) queries the `account_parties` table in the Database (DB) via Postgres connection to confirm that the account exists for the customer.

```
GET
/rest/v1/account_parties?select=account_id&customer_id=eq.b15e81f3-7a2f-4f9
9-bef9-cdeb21464417 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
Accept: application/json
Accept-Profile: public
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW
9oY2NtYXB2cHNwYXVwYmFoIiwicm9sZSI6ImFub24iLCJpYXQiOjE3NDg0ODA2MjQsImV4cCI6M
jA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
Authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJp
c3MiOiJodHRwczovL3ZvbW9oY2NtYXB2cHNwYXVwYmFoLnN1cGFiYXNlLmNvL2F1dGgvdjEiLCJ
zdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhdWQiOiJhdXRoZW
50aWNhdGVkIiwiZXhwIjoxNzU3NDY5MzE0LCJpYXQiOjE3NTc0NjU3MTQsImVtYWlsIjoidGVzd
G5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZGF0YSI6eyJwcm92aWRlciI6ImVt
YWlsIiwicHJvdmlkZXJzIjpbImVtYWlsIl19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGV
zdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92ZXJpZmllZCI6dHJ1ZSwicGhvbmVfdmVyaWZpZWQiOm
ZhbHNlLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZ
SI6ImF1dGhlbnRpY2F0ZWQiLCJhYWwiOiJhYWwxIiwiYW1yIjpbeyJtZXRob2QiOiJwYXNzd29y
ZCIsInRpbWVzdGFtcCI6MTc1NzQ2NTcxNH1dLCJzZXNzaW9uX2lkIjoiYmVhZGNlZjItNjdmMy0
0OWIzLWExZGMtZWJiNjkzYWMyNjBlIiwiaXNfYW5vbnltb3VzIjpmYWxzZX0.uaIL3wqxHooPIP
aLJ_cZx6CGQ8XSCGqn1F3XSmjwvm0
x-client-info: supabase-js-react-native/2.49.5-next.1
```

4. The Client App (CA) then sends another request via REST API to the Web Server (WS) to retrieve account number from the `accounts` table in the Database (DB) via Postgres connection.

```
GET
/rest/v1/accounts?select=*&account_id=eq.4fc4ca5e-09d2-433c-9321-add4c91aa6
a2 HTTP/1.1
Host: vomohccmapvpspaupbah.supabase.co
Accept: application/vnd.pgrst.object+json
Accept-Profile: public
apikey:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InZvbW
9oY2NtYXB2cHNwYXVwYmFoIiwicm9sZSI6ImFub24iLCJpYXQiOjE3NDg0ODA2MjQsImV4cCI6M
jA2NDA1NjYyNH0.kAtm3_e8PUjQGsacDTt7nzrBmQ49H-wU4J4_biPMSZQ
Authorization: Bearer
eyJhbGciOiJIUzI1NiIsImtpZCI6Ik4vampuNklBTGRLeHZpRmUiLCJ0eXAiOiJKV1QifQ.eyJp
c3MiOiJodHRwczovL3ZvbW9oY2NtYXB2cHNwYXVwYmFoLnN1cGFiYXNlLmNvL2F1dGgvdjEiLCJ
zdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDciLCJhdWQiOiJhdXRoZW
50aWNhdGVkIiwiZXhwIjoxNzU3NDY5MzE0LCJpYXQiOjE3NTc0NjU3MTQsImVtYWlsIjoidGVzd
G5pY0BnbWFpbC5jb20iLCJwaG9uZSI6IiIsImFwcF9tZXRhZGF0YSI6eyJwcm92aWRlciI6ImVt
YWlsIiwicHJvdmlkZXJzIjpbImVtYWlsIl19LCJ1c2VyX21ldGFkYXRhIjp7ImVtYWlsIjoidGV
zdG5pY0BnbWFpbC5jb20iLCJlbWFpbF92ZXJpZmllZCI6dHJ1ZSwicGhvbmVfdmVyaWZpZWQiOm
ZhbHNlLCJzdWIiOiI2NmMwMzUwNS05ZDZmLTRkNTAtYjAyZC00MjRlYWYxNjExZDcifSwicm9sZ
SI6ImF1dGhlbnRpY2F0ZWQiLCJhYWwiOiJhYWwxIiwiYW1yIjpbeyJtZXRob2QiOiJwYXNzd29y
ZCIsInRpbWVzdGFtcCI6MTc1NzQ2NTcxNH1dLCJzZXNzaW9uX2lkIjoiYmVhZGNlZjItNjdmMy0
0OWIzLWExZGMtZWJiNjkzYWMyNjBlIiwiaXNfYW5vbnltb3VzIjpmYWxzZX0.uaIL3wqxHooPIP
aLJ_cZx6CGQ8XSCGqn1F3XSmjwvm0
x-client-info: supabase-js-react-native/2.49.5-next.1
```

5. The Client App (CA) encodes the account number as a string into a QR code, and displays it on screen for others to scan

Note: the logical data flow should be user logged in then generate a QR code. The flow is as such to illustrate the information needed to generate the QR code.